

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 1- INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA1205	Course Title:	Computer Architecture for Multicore Systems
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2-INDUSTRY PROBLEM SOLVING TASK
Weightage:	30%	Total Marks:	25
CO4 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
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Department:	CSE(AI)	Date:	07-08-026

ANNEXURE A
INDUSTRY PROBLEM SOLVING TASK

1. Smartphone Performance Optimization
A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.
2. Cloud Server Memory Bottleneck
A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.
3. Autonomous Vehicle Data Processing
An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.
4. AI Inference Edge Device
An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.
5. High-Performance Gaming Console
A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

Code of Conduct:

I SHAIK HAZIPEERA (192572056) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

S. Hazipeera

Signature of the student:

MARKS ALLOCATION

	Understanding of Industry Problem (20%)	Application of Engineering Concepts (25%)	Solution Design (25%)	Use of Tools (15%)	Analysis (10%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

ANSWERS

1. Smartphone Performance Optimization

Problem

A smartphone experiences lag while switching between apps because the processor has to fetch data from slower memory (DRAM) when it is not available in the cache.

Role of Memory Components

L1 Cache

- Smallest and fastest cache located inside the CPU core.
- Stores frequently used instructions and data.
- Provides the quickest access to active applications.

L2 Cache

- Larger than L1 but slightly slower.
- Stores recently used data that cannot fit in L1.
- Reduces CPU waiting time and improves performance.

L3 Cache

- Shared among all CPU cores.
- Stores common data used by multiple applications.
- Improves multitasking and speeds up app switching.

DRAM

- Main memory that stores all running applications and the operating system.
- Has much larger capacity but higher latency than cache.
- Used when the required data is not available in L1, L2, or L3 cache.

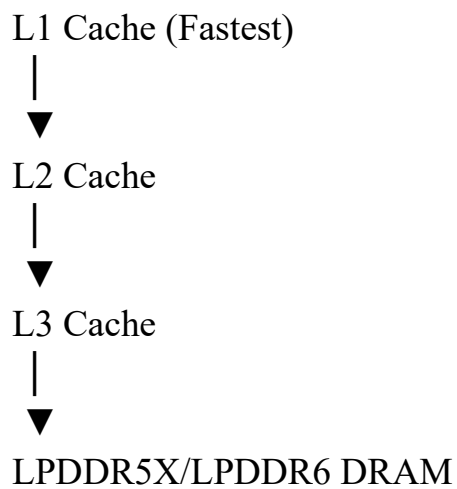
Proposed Hierarchical Memory Redesign

- Increase the size of **L2 and L3 caches** to store more application data.
- Use **LPDDR5X/LPDDR6 DRAM** for higher bandwidth and lower latency.
- Implement **cache prefetching** to load frequently used apps into cache before they are opened.
- Use an **efficient cache replacement policy** to reduce cache misses during multitasking.

Memory Hierarchy

CPU





Benefits

- Faster app switching.
- Reduced memory access latency.
- Higher CPU throughput.
- Better multitasking performance.
- Lower power consumption due to fewer DRAM accesses.

2. Cloud Server Memory Bottleneck

Problem

A cloud service provider experiences high latency in database queries. Compare **cache hierarchy** and **virtual memory paging**, and recommend improvements using appropriate memory technologies.

Cache Hierarchy vs Virtual Memory Paging

Cache Hierarchy	Virtual Memory Paging
Uses L1, L2, and L3 cache.	Uses RAM and disk (SSD/HDD).
Very fast memory access.	Slower due to page swapping.
Stores frequently accessed data.	Stores inactive pages when RAM is full.
Reduces CPU access time.	Increases access time during page faults.
Improves database query speed.	Can reduce performance if excessive paging occurs.

Cause of High Latency

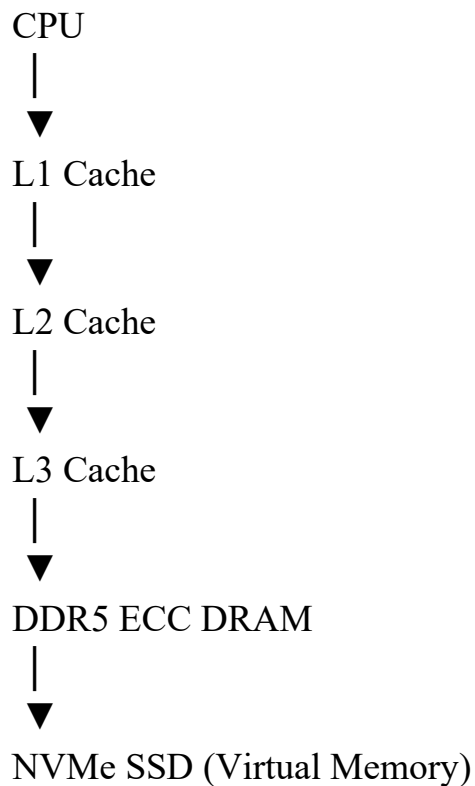
- Database queries frequently access large amounts of data.
- If the required data is not present in the cache, the CPU accesses DRAM.

- When RAM becomes full, virtual memory moves data between RAM and SSD (paging), causing **page faults** and increasing latency.

Recommended Improvements

- Increase the size of **L3 cache** to reduce cache misses.
- Upgrade to **high-speed DDR5 ECC DRAM** for better bandwidth and reliability.
- Increase RAM capacity to minimize virtual memory paging.
- Use **NVMe SSDs** instead of HDDs for faster page swapping.
- Implement intelligent cache management to keep frequently accessed database records in memory.

Optimized Memory Hierarchy:



Benefits

- Faster database query execution.
- Reduced cache misses and page faults.
- Lower memory access latency.
- Improved server throughput.
- Better reliability and scalability for cloud applications.

3. Autonomous Vehicle Data Processing

Problem

An autonomous vehicle must process sensor data in real time. Evaluate **SRAM, DRAM, and MRAM** and design a memory hierarchy that maximizes bandwidth and minimizes latency.

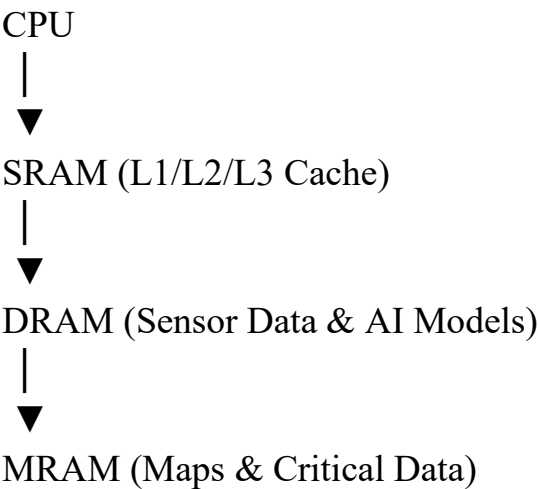
Comparison of Memory Technologies

Memory	Features	Role in Autonomous Vehicle
SRAM	Fastest, low latency, expensive, small capacity	Used as CPU cache for real-time sensor processing.
DRAM	Large capacity, moderate speed, volatile	Stores operating system, AI models, and sensor data.
MRAM	Non-volatile, fast, low power, high endurance	Stores critical data and enables fast system startup.

Memory Hierarchy Design

- **SRAM (L1/L2/L3 Cache):** Stores frequently accessed sensor and control data for instant processing.
- **DRAM:** Holds large volumes of camera, LiDAR, radar, and AI processing data.
- **MRAM:** Stores navigation maps, vehicle settings, and safety-critical information that must be retained even after power loss.

Optimized Memory Hierarchy



Recommended Improvements

- Increase **SRAM cache** size to reduce memory access time.
- Use **high-bandwidth DDR5/LPDDR5 DRAM** for faster sensor data transfer.
- Integrate **MRAM** for storing critical data with low power consumption.
- Optimize memory scheduling to prioritize real-time sensor processing.

Benefits

- Faster real-time decision making.
- Reduced memory latency.
- High bandwidth for processing camera and LiDAR data.
- Lower power consumption.
- Improved reliability and data retention during power interruptions.

4. AI Inference Edge Device

Problem

An edge AI device needs **fast model loading** with **low power consumption**. Compare **NAND Flash, DRAM, and RRAM** and recommend an optimized hierarchical memory structure.

Comparison of Memory Technologies

Memory	Features	Role in Edge AI Device
NAND Flash	Non-volatile, high storage capacity, slower speed	Stores AI models, operating system, and application files.
DRAM	High speed, volatile, medium capacity	Loads AI models into memory for execution and processing.
RRAM	Non-volatile, very fast, low power consumption	Stores frequently used AI models and parameters for quick access.

Memory Hierarchy Design

- **NAND Flash:** Stores the operating system, AI models, and application data permanently.
- **DRAM:** Loads the AI model from NAND Flash and provides fast access during inference.

- **RRAM:** Stores frequently used AI models and parameters to reduce loading time and power consumption.

Optimized Memory Hierarchy

CPU/AI Accelerator



DRAM



RRAM



NAND Flash

Recommended Improvements

- Use **RRAM** as a fast intermediate memory for frequently used AI models.
- Upgrade to **LPDDR5 DRAM** for higher bandwidth and lower latency.
- Store large AI models in **NAND Flash** and load only required parts into DRAM.
- Implement intelligent caching and model prefetching to reduce loading time.

Benefits

- Faster AI model loading.
- Lower power consumption.
- Reduced latency during inference.
- Improved processing speed.
- Better battery life for edge devices.

5. High-Performance Gaming Console

Problem

A gaming console suffers **frame drops during large scene loading**. Analyze **L3 cache vs DRAM throughput** and propose memory hierarchy enhancements.

Comparison of L3 Cache and DRAM

L3 Cache	DRAM
Shared among all CPU cores.	Main memory of the system.
Very low latency and high-speed access.	Higher latency but much larger capacity.
Stores frequently used game data.	Stores game textures, maps, and assets.
Improves CPU performance and reduces delays.	Provides high bandwidth for loading large game files.

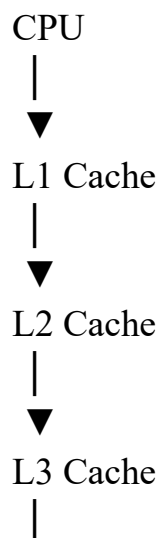
Cause of Frame Drops

- Large game scenes require loading textures, models, and maps.
- If data is not available in the **L3 cache**, the CPU accesses **DRAM**, which has higher latency.
- Frequent cache misses and heavy DRAM access reduce frame rate and cause frame drops.

Memory Hierarchy Enhancements

- Increase the **L3 cache size** to store more frequently used game data.
- Upgrade to **GDDR6/GDDR6X or DDR5 DRAM** for higher bandwidth.
- Use **cache prefetching** to load upcoming game assets before they are needed.
- Optimize cache management to reduce cache misses during gameplay.

Optimized Memory Hierarchy





GDDR6/GDDR6X DRAM

Advantages

- Reduced frame drops.
- Faster scene and texture loading.
- Lower memory access latency.
- Higher frame rate and smoother gameplay.
- Improved overall gaming performance.